

Project Rubrics

GitHub Secrets

Configured in:

Settings → **Secrets and Variables** → **Actions**

Required:

- AWS_ACCESS_KEY_ID
 - AWS_SECRET_ACCESS_KEY
 - AWS_SESSION_TOKEN
 - AWS_REGION
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Pipeline Phases

Phase 1 — Testing

Students should:

- Run unit and integration tests
 - Generate test reports
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Phase 2 — Infrastructure Provisioning (Terraform)

Students must:

- Initialize Terraform
- Validate configuration
- Plan and apply infrastructure

Required S3 configuration:

- Unique bucket name
- Versioning enabled
- Encryption enabled
- Public access blocked

Phase 3 — Container Build and ECS Deployment

Students must:

- Build Docker image
- Push image to ECR
- Deploy service to ECS Fargate
- Verify service is running

Dockerfile requirements:

- Multi-stage build
- Non-root user
- Healthcheck configured

OR

Phase 4 — Kubernetes Deployment (EKS)

Students must:

- Configure cluster access
- Deploy Kubernetes manifests
- Expose service endpoint
- Ensure deployment stability

Kubernetes requirements:

- Minimum 2 replicas
 - Resource limits defined
 - Liveness and readiness probes
 - Non-default namespace
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Workflow Order

Push / PR



Run Tests



Terraform Apply



Docker Build & Push



Deploy to ECS

or

Deploy to EKS